

SYLLABUS :-

Course Overview:This course gives an overall understanding of design, drafting, modelling and rendering software that are essential for later design studios and architectural and management projects. **Learning Objectives:**1. To understand Design and drafting software and applications2. To understand presentation software and applications3. To understand modelling software like BIM and applications4. To understand evaluation software like Climate Consultant and applications5. To understand management software like Primavera and applications

Methodology: Structured lectures, examples and discussions.

Projects on assigned buildings and practical sessions and working on computers in lab. Being a practice oriented course, students are expected to work in continuation without missing practical sessions and lectures. **Course**

Curriculum:Module 1 : Design Drafting using AutoCAD; setting of limits and units, line, poly line, shapes point and wall drafting. Use of array and spline and other drafting tools. Positioning of columns and services and integration through the use of construction lines, and Grid. Lettering and dimensioning and use of - 23 -18 | Department of Architecture and Regional Planning, IIT Kharagpur layers. Sheet composition and plotting styles. Use of block and world block. Construction of library of furniture landscape elements and hatching.

Presentation and submission formalities included in plotting as per BIS. 3D modelling using AutoCAD and its applications.Module 2 : Presentation software and its applications; Sketch-Up, Introduction to Adobe Photoshop, Illustrator and modelling software like Grasshopper, with its examples and application. Use and application of Climate consultant in designs and decision making.Module 3 : Application and use of REVIT and production of design and integration using levels and planes, grid and numbering. Designing and drafting walls and columns, Dimension styles, lettering, positioning of doors windows, atrium and balconies. Use and formation of library and generation of 3D and walkthrough using camera and setting. Landscapes and grades representation, positioning of landscape elements. Designing and construction of multi-layered building elements. Basic estimation of quantities. Composing sheets and plotting according to BIS.Module 4 : Introduction to BIM and Primavera for Management of construction projects. Coordinating multiple services and conflict resolution in designs. Study of sample projects and its assessment.**Reading List:**1.

Reference manual of REVIT and AutoCAD of the appropriate version taught.2.

Reference manual of Adobe Photoshop, Illustrator, Flash3. Reference manual of PRIMAVERA4. BIS stands for production of Drawings